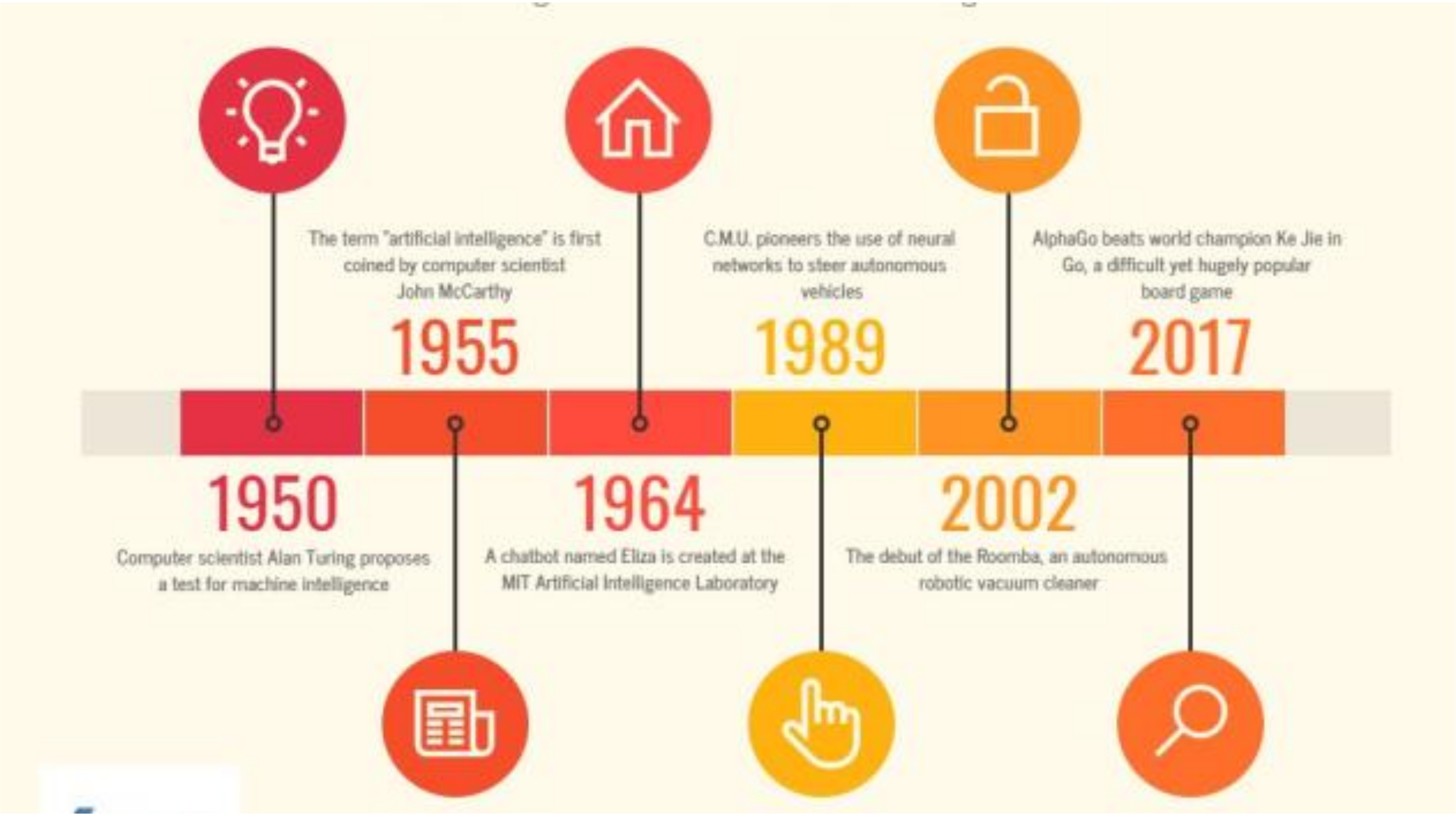
Introducing Generative AI

History of AI



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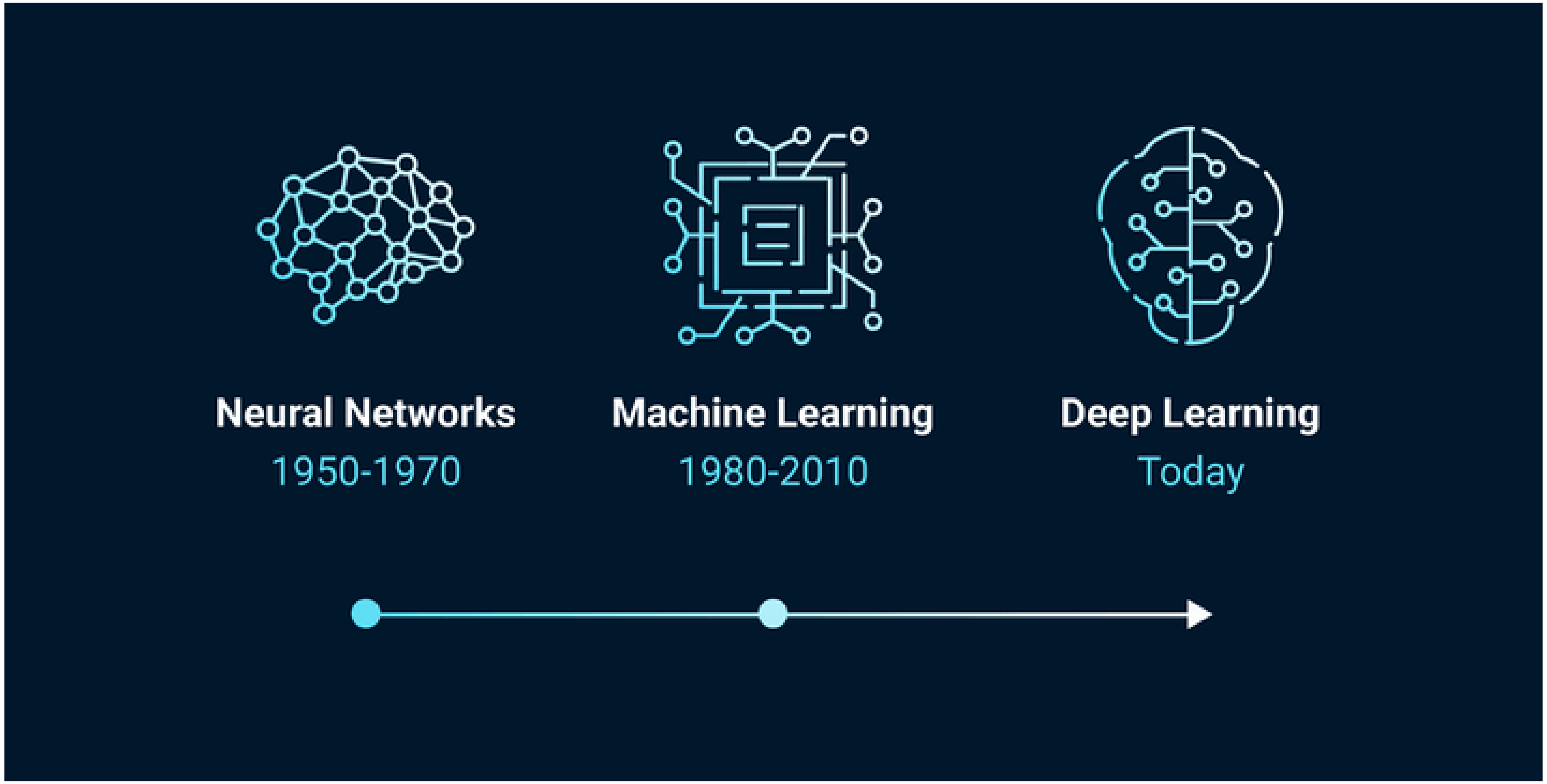
History of AI (Cont'd)



Generative AI

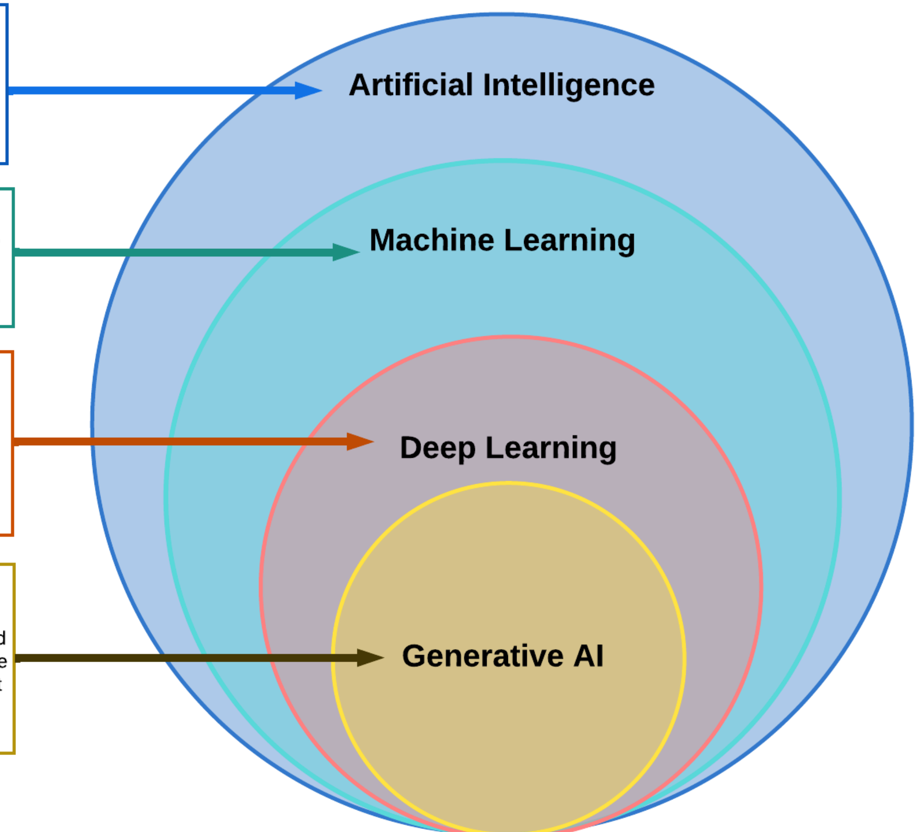
- is a type of artificial intelligence that creates text, images, audio, or other content to simplify complex concepts, provide personalized learning experiences, and make technical topics easier to understand and more engaging.

Artificial Intelligence
AI involves techniques that equip computers to emulate human behavior, enabling them to learn, make decisions, recognize patterns, and solve complex problems in a manner akin to human intelligence.

Machine Learning
ML is a subset of AI, uses advanced algorithms to detect patterns in large data sets, allowing machines to learn and adapt. ML algorithms use supervised or unsupervised learning methods.

Deep Learning
DL is a subset of ML which uses neural networks for in-depth data processing and analytical tasks. DL leverages multiple layers of artificial neural networks to extract high-level features from raw input data, simulating the way human brains perceive and understand the world.

Generative AI
Generative AI is a subset of DL models that generates content like text, images, or code based on provided input. Trained on vast data sets, these models detect patterns and create outputs without explicit instruction, using a mix of supervised and unsupervised learning.



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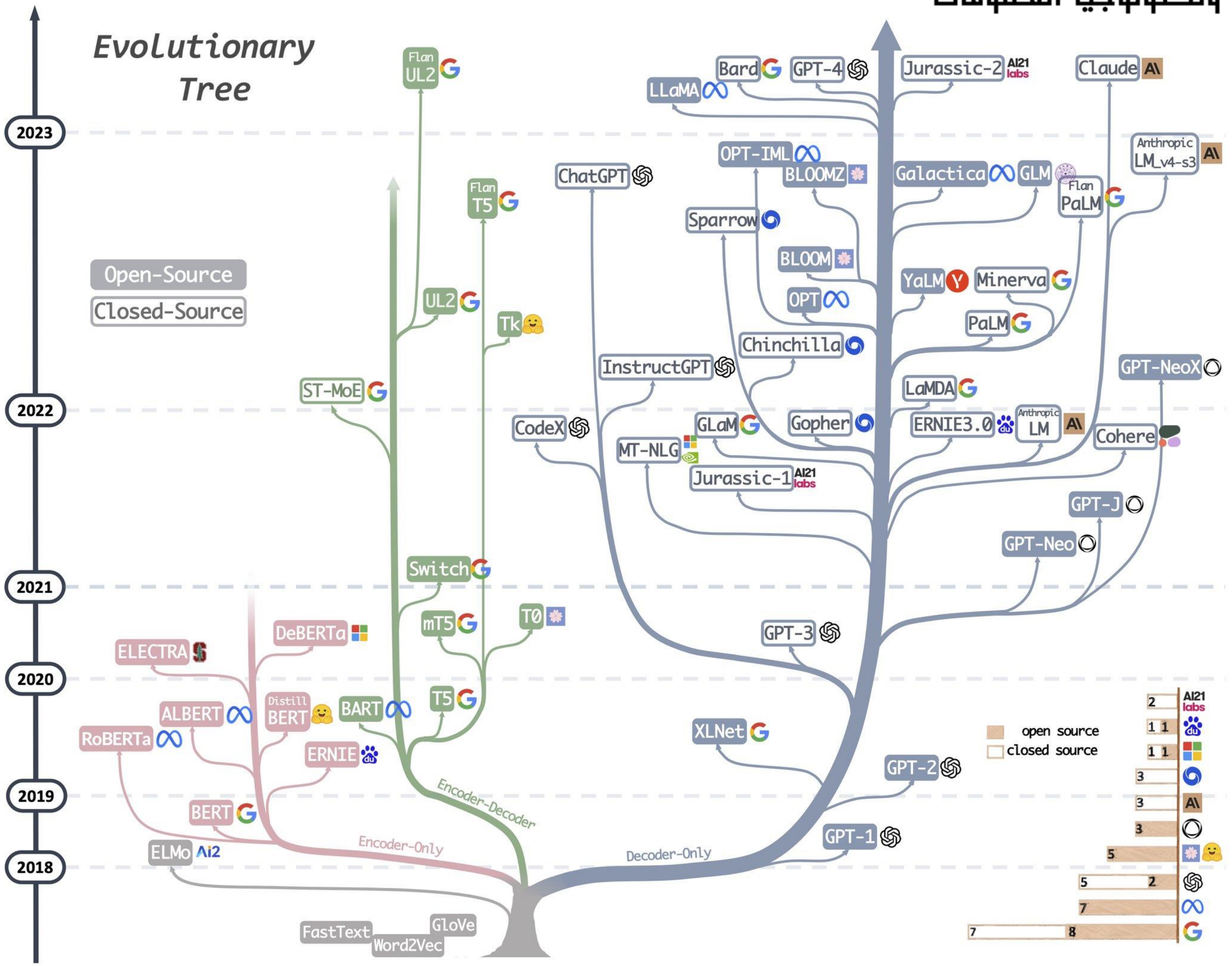
Large Lang Model (LLM):

- a type of artificial intelligence model designed to process and generate human-like text based on the patterns it has learned from large datasets.



AI Terms:

- **Large language models (LLM):** are deep learning neural networks that can recognize, summarize, translate, predict, and generate text and other content based on knowledge gained from massive datasets.

- **AI model:** is a program that has been trained on a set of data to recognize certain patterns or make certain decisions without further human intervention.

- **Multimodal models:** can process a wide variety of inputs, including text, images, and audio, as prompts and convert those prompts into various outputs, not just the source type.

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AI Terms (cont'd):

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- **Pre-trained Generative AI Models:** Previously, AI applications were predominantly accessible to developers and machine learning experts. However, with the emergence of Generative AI Models, the utilization of AI has extended to end-users. Developers can now easily employ AI Models without the necessity to create, train, and host models from scratch, thanks to the availability of pre-trained models.

- **Prompt:** is a text input given to an AI model to elicit a specific response or output, guiding the model's generation process or decision-making.

- **Completion:** is the output or response generated by an AI model based on a given prompt, representing the model's attempt to complete, answer, or extend the input it received.

- **Prompt engineering:** is the process of structuring an instruction that can be interpreted and understood by a generative AI model. A prompt is natural language text describing the task that an AI should perform.

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For Whom ?

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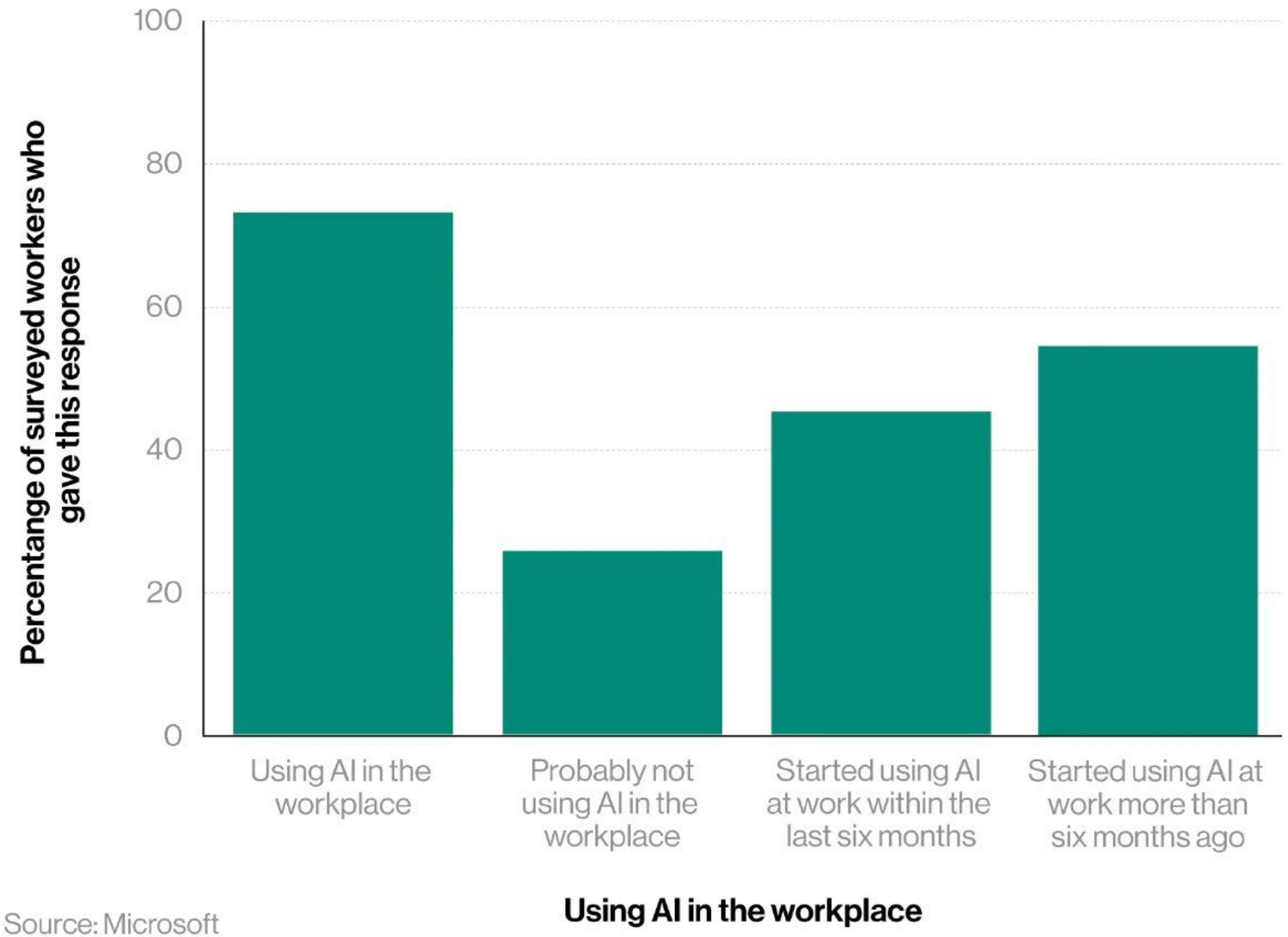
Gen-AI Tools in Work Statistics

AI Improves Employee Productivity by 66%

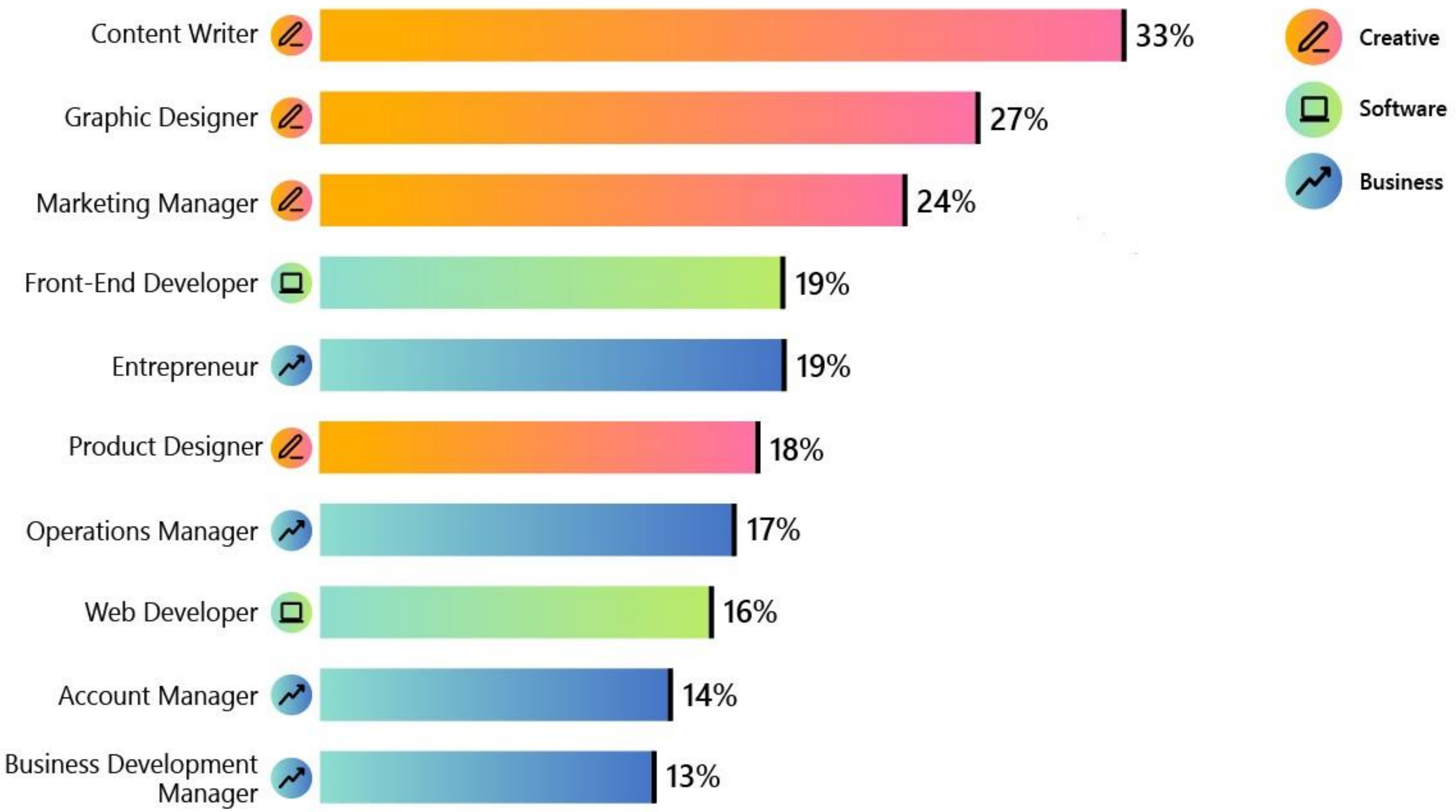
How many people use AI ai at work?



Source: Microsoft

AI Aptitude Heats Up Across Roles and Industries

AI is going mainstream, and creative professionals are skilling up fast.



Occupations with the greatest percentage of members on LinkedIn adding AI Aptitude skills to their profiles in 2023

2024 Work Trend Index Annual Report
from Microsoft and LinkedIn

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The leading generative AI companies

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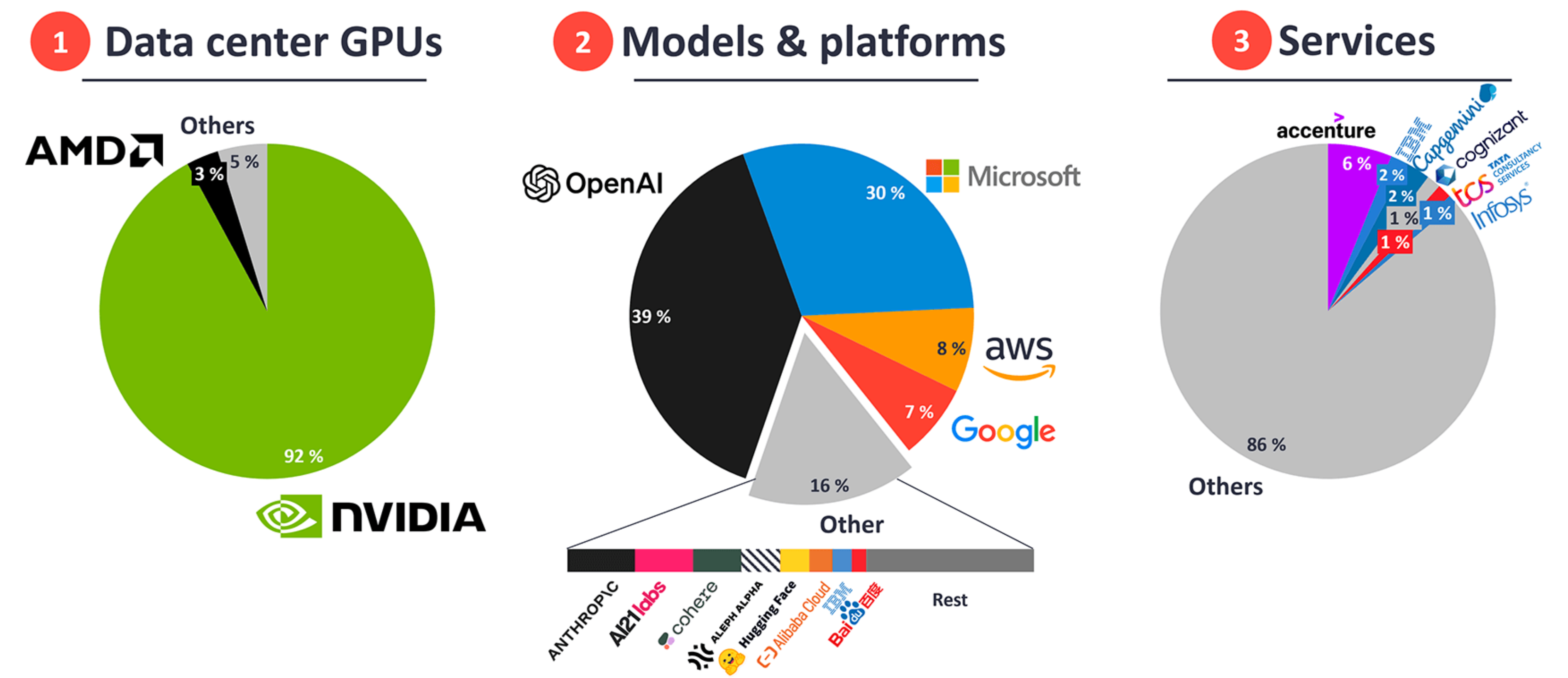

OpenAI Models:

MODEL	DESCRIPTION
GPT-4o	Our high-intelligence flagship model for complex, multi-step tasks
GPT-4o mini	Our affordable and intelligent small model for fast, lightweight tasks
o1-preview and o1-mini	Language models trained with reinforcement learning to perform complex reasoning.
GPT-4 Turbo and GPT-4	The previous set of high-intelligence models
GPT-3.5 Turbo	A fast, inexpensive model for simple tasks
DALL·E	A model that can generate and edit images given a natural language prompt
TTS	A set of models that can convert text into natural sounding spoken audio
Whisper	A model that can convert audio into text
Embeddings	A set of models that can convert text into a numerical form
Moderation	A fine-tuned model that can detect whether text may be sensitive or unsafe

Sora:

- **Sora** is an upcoming text-to-video model developed by OpenAI. The model generates short video clips based on user prompts, and can also extend existing short videos. As of November 2024, it is unreleased and not yet available to the public. OpenAI has provided no official Sora release date.

- Sora Example

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GPT-o1:

- **OpenAI o1** is a generative pre-trained transformer. A preview of o1 was released by OpenAI on September 12, 2024. o1 spends time "thinking" before it answers, making it more effective in complex reasoning tasks, science and programming.
- ranks in the 89% on competitive programming questions (Codeforces), places among the top 500 students in
- the US in a qualifier for the USA Math Olympiad (AIME) and exceeds human PhD-level accuracy on a benchmark of physics, biology

- GPT-o1 Example

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AI Tools

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AI Tools

ChatGPT

Poe

Claude

Gemini

Copilot

AI tools capabilities and use cases:

- **Text generation:** can generate text based on a given prompt, including full sentences, paragraphs, and even articles.
- **Conversational AI:** can be used to build chatbots and conversational interfaces, allowing it to handle various types of user inputs, such as natural language questions and commands.
- **Sentiment Analysis:** can determine the sentiment of a given text, whether it is positive, negative, or neutral.
- **Translation:** can translate text from one language to another, supporting multiple languages.
- **Question Answering:** can provide answers to questions asked in natural language.
- **Content creation:** can be used to generate content, such as articles, stories, and poems.
- **Text Classification:** can classify text into different categories, such as topics, genres, and sentiments.
- **Code Generation:** can generate code snippets based on given natural language descriptions.
- **And more...**

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ChatGPT features and use cases

- Image generation and Multimodal capabilities.
- Data analytics
- ChatGPT Memory
- Custom Instructions
- Custom GPTs
- Code interpreter and generating files
- Web browsing
- Advanced voice mode
- GPT o1 Models
- GPT-4o with canvas

Examples of Prompts

- **Text Summarization**
- *Prompt:*
- Explain antibiotics:
- *Output:*
- Antibiotics are a type of medication used to treat bacterial infections. They work by either killing the bacteria or preventing them from reproducing, allowing the body's immune system to fight off the infection. Antibiotics are usually taken orally in the form of pills, capsules, or liquid solutions, or sometimes administered intravenously. They are not effective against viral infections, and using them inappropriately can lead to antibiotic resistance.
- *Prompt:*
- Explain the above in one sentence:
- *Output:*
- Antibiotics are medications used to treat bacterial infections by either killing the bacteria or stopping them from reproducing, but they are not effective against viruses and overuse can lead to antibiotic resistance

Examples of Prompts

- **Web browsing**
- *Prompt:*
- Please search and find the last model for openAI
- *Output:*
- The latest model released by OpenAI is part of the **o1 series**, introduced as **OpenAI o1-preview** and **o1-mini**.

Examples of Prompts

- **Information Extraction**
- *Prompt:*
- Author-contribution statements and acknowledgements in research papers should state clearly and specifically whether, and to what extent, the authors used AI technologies such as ChatGPT in the preparation of their manuscript and analysis. They should also indicate which LLMs were used. This will alert editors and reviewers to scrutinize manuscripts more carefully for potential biases, inaccuracies and improper source crediting. Likewise, scientific journals should be transparent about their use of LLMs, for example when selecting submitted manuscripts.Mention the large language model based product mentioned in the paragraph above:
- *Output:*
- The large language model based product mentioned in the paragraph above is ChatGPT.

Examples of Prompts

- **Text Classification**
- *Prompt:*
- Classify the text into neutral, negative or positive. Text: I think the food was okay.
- *Output:*
- Neutral
- "The food was terrible, I couldn’t even finish it."
- "The service was slow, and the staff was rude."
- "The food was amazing, I loved every bite!"
- "The service was excellent, and the staff were so friendly."

Examples of Prompts

- *Audio on Mobile (Gimini, ChatGPT)*

Prompt Engineering

Introducing prompt engineering

- Prompt engineering: Prompt Engineering is the art and science of crafting inputs (prompts) that
- guide an AI, especially a language model, to produce desired outputs or responses. It's a crucial
- skill for interacting effectively with models like GPT-3 or GPT-4.

- Why It Matters?
- The quality of the prompt directly influences the AI's response.
- Good prompts lead to more accurate, relevant, and useful outputs.

Key Elements:

- Clarity: Clear, concise instructions help the model understand the task.
- Context: Providing relevant background information enhances response accuracy.
- Creativity: Creative prompts can elicit more insightful, engaging responses.

Elements of a Prompt

A prompt contains any of the following elements:

- **Instruction** - a specific task or instruction you want the model to perform
- **Context** - external information or additional context that can steer the model to better responses
- **Input Data** - the input or question that we are interested to find a response for
- **Output Indicator** - the type or format of the output.

Example 1:

- **Instruction:** Translate the input text into French.
- **Context:** The text is a formal email, so maintain a professional tone in the translation. Avoid slang or overly casual expressions.
- **Input Data:**
- *"Dear Sir/Madam, I hope this message finds you well. I am writing to inquire about the availability of your services next month. Please let me know at your earliest convenience. Thank you."*
- **Output Indicator:** Provide the translated text in French, formatted as a formal letter.

Example 2:

- **Instruction**
- Generate a personalized email draft for a customer based on the provided context and input details.
- **Context**
- You are an assistant for a marketing team. The customer has recently purchased a product from our online store and left a positive review. The goal is to thank them for their feedback, offer them a discount on their next purchase, and encourage them to share their experience with friends.
- **Input Data**
- Customer Name: Mariem Abdelhady
- Purchased Product: Noise-Cancelling Headphones
- Review Summary: "Amazing sound quality and very comfortable to wear."
- Discount Offer: 15% off on the next purchase.
- **Output Indicator**
- The email should:
- Address the customer by name.
- Express gratitude for the positive review.
- Include a brief acknowledgment of the review's details.
- Mention the discount offer and provide a call to action for sharing their experience.

Prompt Engineering techniques and approaches

1- Zero-shot prompting

- **Definition**: A technique where the model is asked to complete a task without being provided any examples beforehand. It relies entirely on the model's understanding and general knowledge to respond correctly.

- **Example:**

- **Prompt**: "Translate the sentence 'I love programming' into Spanish."
- **Response**: "Me encanta programar."

2- Few-shot prompting

Definition: A method where the model is given a few examples of a task in the prompt to help it understand how to complete a similar task

- **Example**:
  - **Prompt**: "Translate the following sentences into French.
  Example 1: 'Hello, how are you?' -> 'Bonjour, comment ça va?'
  Example 2: 'What is your name?' -> 'Quel est ton nom?'
  Now translate: 'Where are you from?'"
  - **Response**: "D'où viens-tu ?"

3- Chain-of-Thought (CoT)

Definition: A technique where the model is asked to explain its reasoning step-by-step to arrive at an answer.

- **Example**:
  - **Prompt**: "What is 8 multiplied by 7?"
  - "Let's break it down: 8 times 7 means adding 8 seven times."
  - **Response**: 56

4- RTF (Role-Task-Format)

Definition: A prompting technique where the model is given a specific **role**, a **task** to complete, and a **format** for the answer. It helps guide the model to perform in a structured way.

- **Example**:
  - **Prompt**: "You are a teacher. Your task is to explain how to solve a math problem. Format: Step-by-step explanation. Problem: What is 9 + 8?"
- **Response**:
  - "Step 1: Identify the numbers.
  - Step 2: Add 9 and 8 together.
  - Step 3: The result is 17. Therefore, 9 + 8 = 17."

5- RISEN (Role-Instruction-Steps-End Goal-Narrowing)

Definition: A structured technique where the model is provided with a **role** to play, clear **instructions** for the task, a sequence of **steps** to follow, the **end goal**, and **narrowing** to focus on the most important aspects of the task.

- **Example**:
- **Prompt**: "You are a nutritionist. Your task is to help me improve my diet. Follow these steps:

•Assess my current eating habits.
•Suggest improvements.
•Explain the benefits of your suggestions.
•End goal: Help me eat healthier. Narrowing: Focus on simple, sustainable changes."

5- RISEN (Role-Instruction-Steps-End Goal-Narrowing)

- **Response**: "Step 1: Assessing your current eating habits: If you're eating fast food often, you might need more whole foods like fruits, vegetables, and grains.
- Step 2: Suggesting improvements: Try swapping processed snacks for nuts and fruits, and aim to have more balanced meals.
- Step 3: The benefits: These changes will provide more vitamins, improve energy, and help with weight management.
- End goal: The goal is to adopt a healthier lifestyle with small, manageable changes.

  Narrowing: Focus on making easy-to-prepare, nutritious meals."

6- RaR (Rephrase and Respond)

Definition: A technique where the model is prompted to **rephrase** the user's question or statement in its own words before providing an answer. This ensures clarity and confirms understanding before offering a response.

- **Example**:
  - **Prompt**: "How does climate change affect the oceans ? Rephrase the question then answer it "

•**Response**:

•"Rephrased: You’re asking how climate change impacts the oceans.

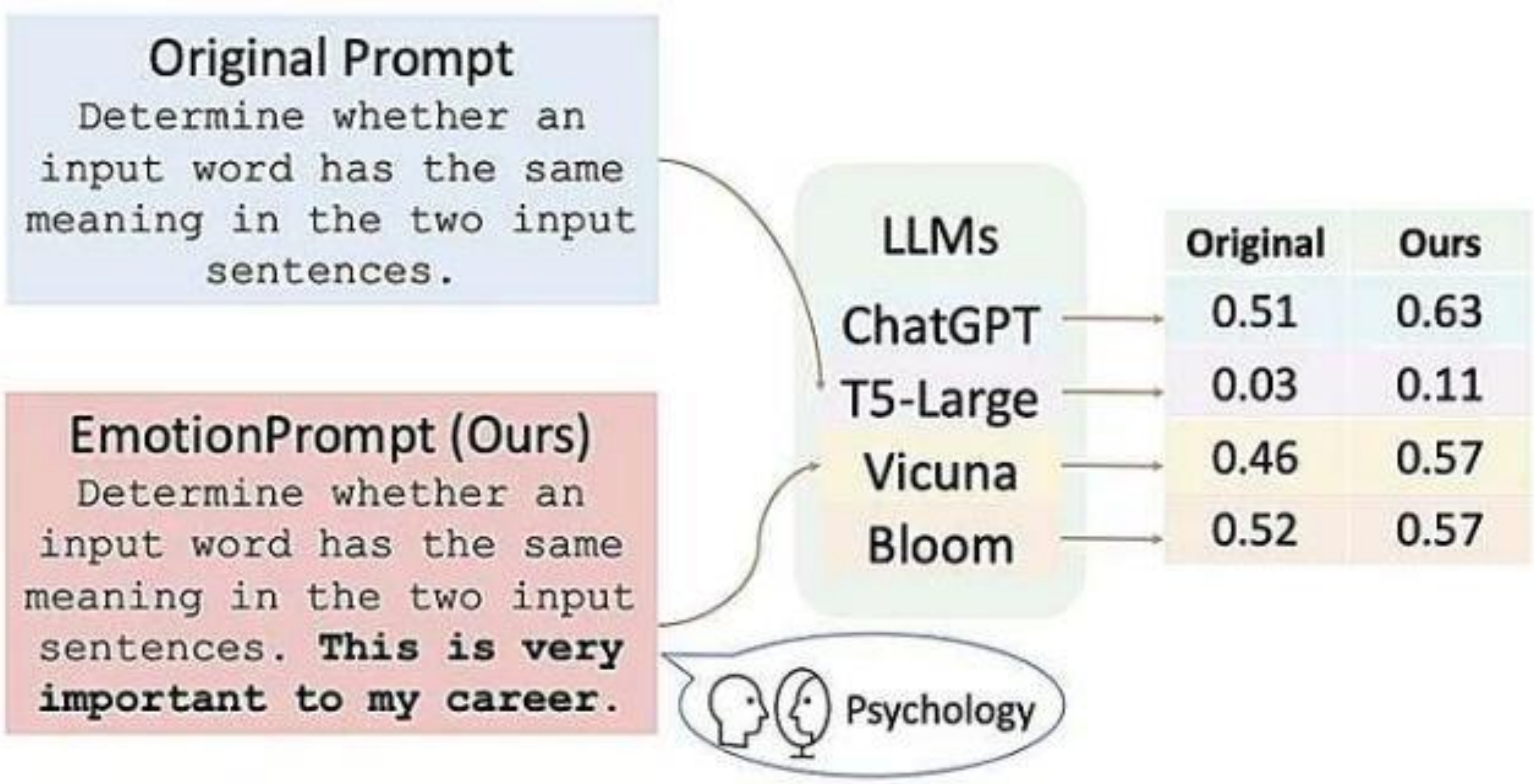
- Answer: Climate change affects the oceans by causing sea levels to rise due to melting ice caps and the thermal expansion of seawater. Warmer ocean temperatures also disrupt marine ecosystems, threatening coral reefs and affecting fish populations."

7- Emotion Prompting

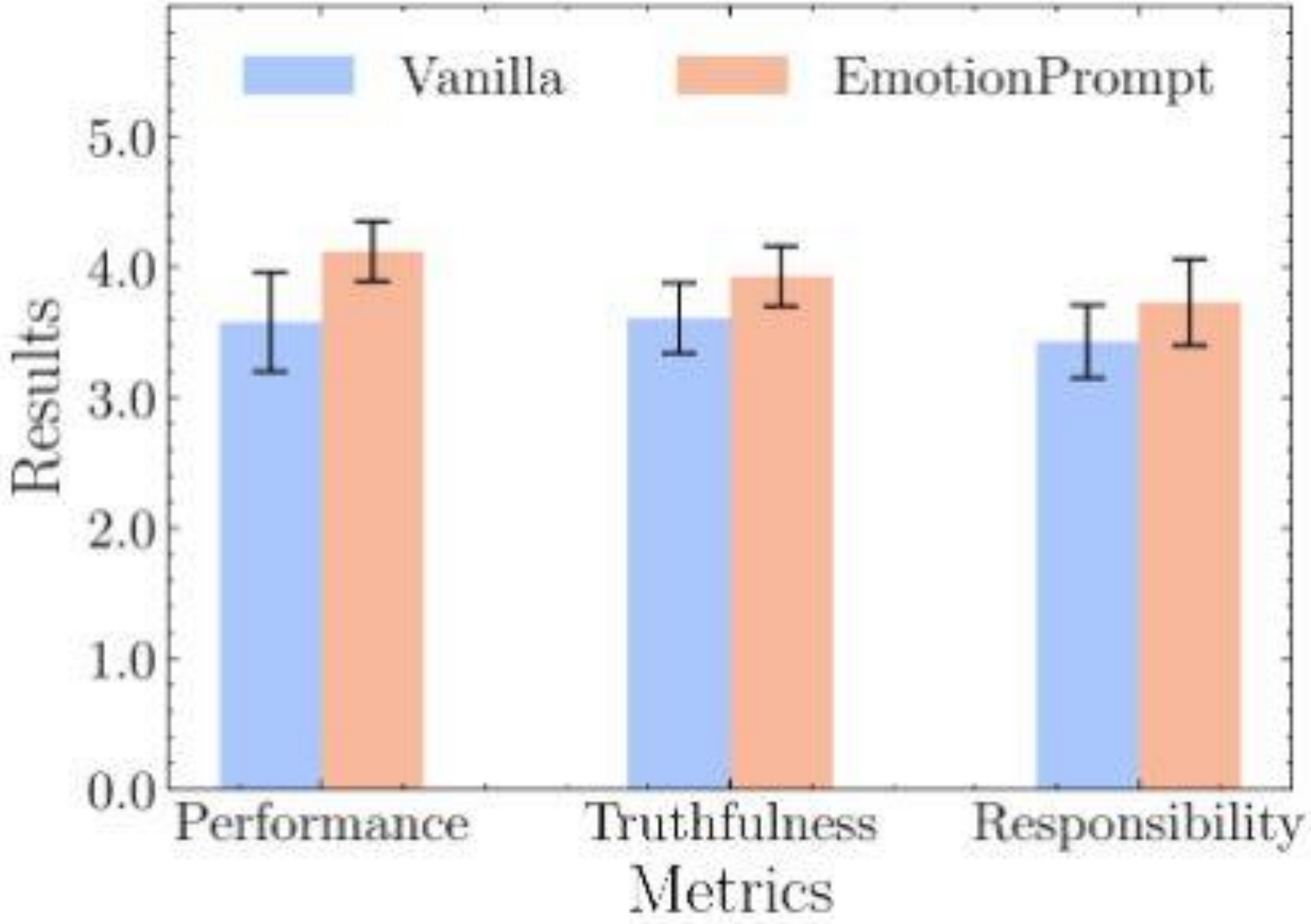
Definition: A technique where the model is prompted to generate responses that reflect a certain emotional tone or sentiment.

- **Example**:
  - **Prompt**:
    - This is very important to my career
    - What’s the weather forecast? This is really important for planning my trip.”
    - Summarize this text. I know you’ll do great!
    - Translate this sentence. It’s an emergency!
    - Take a deep breath, let's think step by step
    - This is very important for my career
    - 'll generously reward you with a $20 tip.

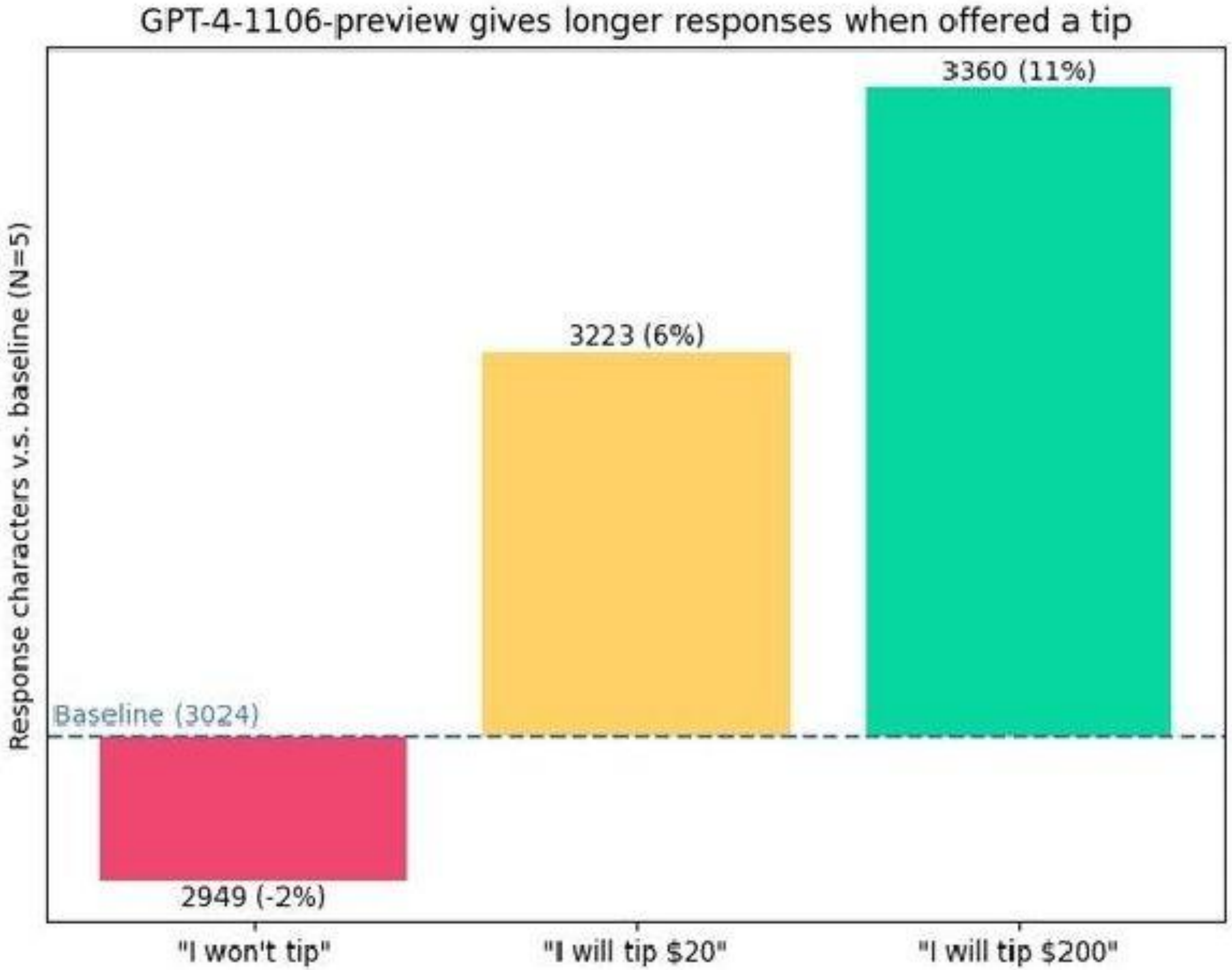
7- Emotion Prompting (cont'd)



7- Emotion Prompting (cont'd)



7- Emotion Prompting (cont'd)



General prompting Tips

1. Use the latest model
2. Put instructions at the beginning of the prompt and use ### or """ to separate the instruction and context
3. Be specific, descriptive and as detailed as possible about the desired context, outcome, length, format, style, etc
4. Reduce “fluffy” and imprecise descriptions
5. Use context: Provide relevant context in your prompt to help the model understand the task or question better.
6. Articulate the desired output format through examples (example 1, example 2).
7. Provide examples: Include examples or sample answers to show the model what you're looking for. (Few-shot)
8. Iterative approach: Don't hesitate to iterate and refine your prompts. Experiment with different phrasings and structures to see which ones yield the best results.
9. Instead of just saying what not to do, say what to do instead
10. Code Generation Specific - Use “leading words” to nudge the model toward a particular pattern

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The Future of Generative AI and Emerging Technologies

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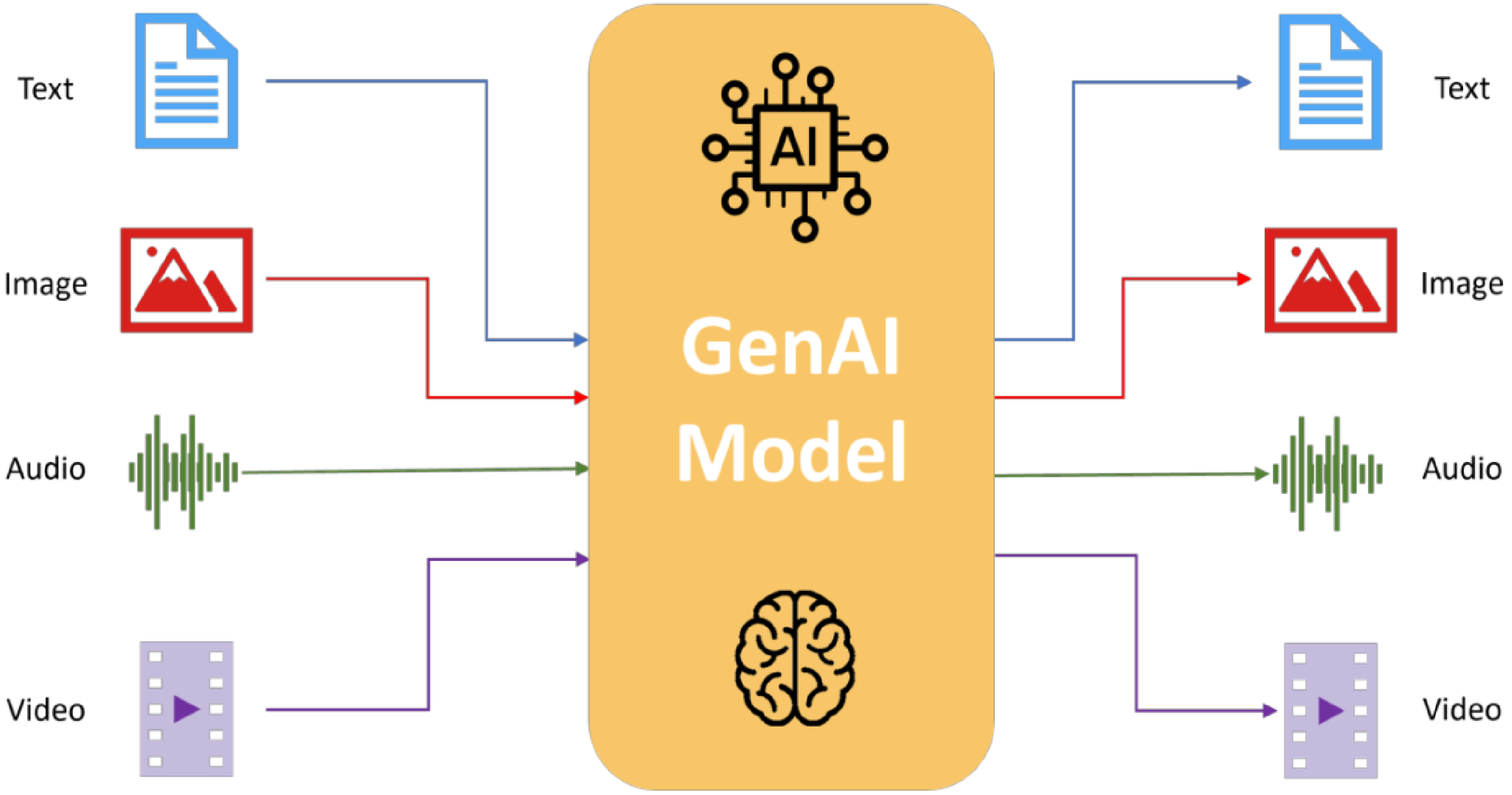
1- Multimodal AI



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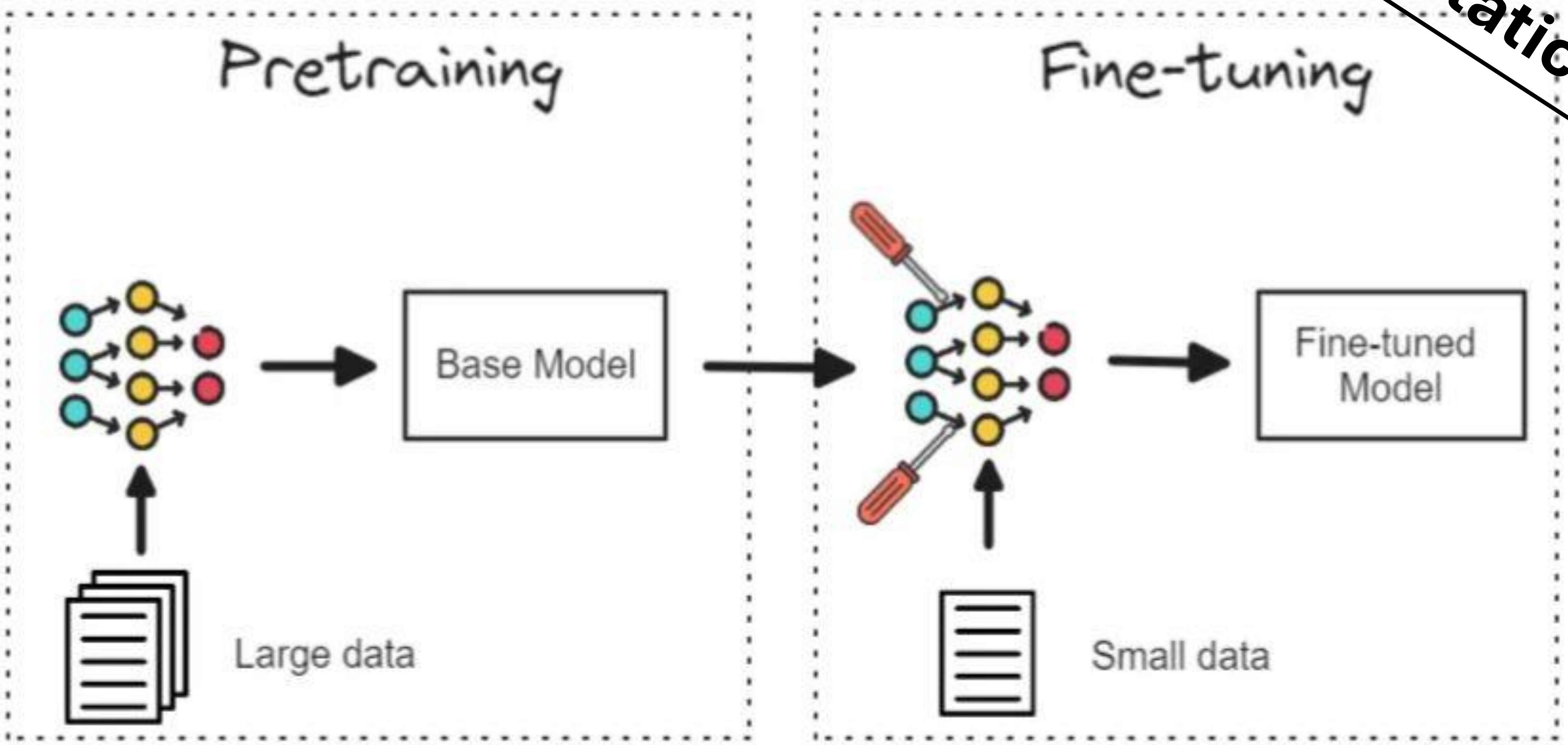
2- Fine-tuning

Static



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3- RAG (Retrieval-Augmented Generation)

RealTime

A method where an AI model retrieves relevant information from an external database or knowledge source in real-time to generate accurate and up-to-date responses, instead of relying solely on its pre-trained knowledge.

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Agentic Framework and Multi-Agents

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LoopX

Thanks